

Predicting post-VR game experiences with wearable physiological sensors[☆]

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ABSTRACT

Players' post-game experiences determine their loyalty to a virtual reality (VR) game. However, methods for identifying players' post-game experiences in the early stages have received far less attention than those for in-game experiences. In this study, we explored the potential of using measurements from wearable physiological sensors to predict players' post-VR game experiences. The methods employed were correlation analyses and machine learning techniques. The results showed that electrodermal activity (EDA) measurements, particularly the mean EDA and mean EDA peak, are associated with players' post-VR game experiences after accounting for noise. By utilizing machine learning technology, physiological metrics can forecast players' diverse reactions after playing VR games with high accuracy. The symptoms of depersonalization/derealization experienced after VR gaming are attributed to being induced by actions within the virtual environment. This research makes significant contributions to the field of user experience recognition and the progression of VR gaming by demonstrating the potential for future VR game centers to analyze player emotions remotely and cost-effectively. This achievement provides the prerequisite for these centers to create tailored new 3D game scenarios to enhance players' post-game experiences with the support of future advanced generative artificial intelligence technologies.

1. Introduction

Virtual reality (VR) represents a fully immersive multimedia experience within a 3D environment, enabling users to engage and interact in scenarios that are either realistic or imagined. In recent years, the VR industry has experienced rapid development. More specifically, global VR revenues are expected to climb from less than US\$16 billion in 2024 to more than US\$18 billion by the end of 2025 [1]. Notably, the VR gaming content industry is forecast to grow from US\$ 1.8 billion in 2020 to US\$ 6.9 billion by 2025 [2]. However, a key factor limiting wider VR adoption is the issue of suboptimal user experiences [3]. The retention of players is critical to the sustainability of the VR game industry [4].

Researchers have dedicated significant effort to understanding players' emotional responses and in-game experiences through a range of methods including observation, physiological monitoring, questionnaires, interviews, focus groups, and expert reviews [5]. In contrast, there has been limited exploration of players' experiences after game sessions. These post-game emotions, often defined as the range of feelings experienced once the game is over [6], can be positive, negative, a

sense of fatigue, or a readjustment to reality [7]. Compared with in-game experience, post-game experience more directly impacts players' attitude and loyalty to the game [8,9,10].

Currently, post-game experience assessments primarily rely on self-reported methods, which can be time-consuming and subjective. A potentially more efficient and objective approach involves the use of physiological data. Recent studies have demonstrated a relationship between physiological measurements and in-VR game experiences, building models to predict various emotional states such as presence, immersion, and flow [11,12]. This present study leverages this previous research, seeking to objectively assess post-VR game experiences through physiological measures.

This study aims to systematically explore the potential of utilizing in-game physiological measurements to predict a range of post-game experiences. The specific objectives are threefold: firstly, to assess the association between physiological measurements and self-reported experiences after VR gaming; secondly, to develop prediction models of post-VR game experiences employing these physiological indicators through machine learning (ML) techniques; and finally, to examine

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whether in-game actions, which can both boost gameplay success and cause motion sickness, have an impact on the symptoms of depersonalization and derealization (DPDR) experienced after VR gaming.

This research offers significant contributions to both the field of user experience recognition and the progression of VR gaming. Particularly, with the advancement of generative artificial intelligence technologies, there is a high potential for 3D games to be created automatically [13,14]. The achievement in this study lays the groundwork for future VR game centers to remotely and non-invasively analyze players' loyalty to their product at a low cost. By utilizing this method, game centers can forecast players' post-game experiences and enable the generative artificial intelligence technologies to create new 3D game scenarios with corresponding adjustments in real time, thereby enhancing players' post-game experiences.

2. Literature review

During VR games, various physiological measures can be obtained through sensors. We chose the E4 wristband [15] and analyzed the collected electrodermal activity (EDA), skin temperature, and movement acceleration data. Compared to other sensors used in affective computing, the E4 wristband is one of the most affordable and quickest-to-wear options in real VR gaming settings. Electroencephalography (EEG), for instance, are more complex, while VR eye-tracking and facial expression recognition tools require modifying the original game programs to track the player's eye movements or facial changes in VR. More details about the E4 wristband and these measures are provided in the Methods section. This section reviews the aforementioned three measures as well as prior research that has connected these measures with user experiences in video games and VR environments. We also review previous research on DPDR, a mental disorder that some players experience following the sudden removal of a VR headset.

2.1. EDA, skin temperature, and movement

EDA is a measure of neurally mediated effects on sweat gland permeability. It reflects the body's response to the central nervous system's integrated attentional, affective, and motivational processes [16]. EDA is characterized by a slowly varying tonic activity and a rapidly varying phasic activity [17]. The tonic level of electrodermal arousal, reflected by the skin conductance level (SCL), can be used to monitor individual variations in the constancy of diurnal rhythms as well as the overall level of arousal [16]. Phasic electrodermal responses, such as the skin conductance response (SCR), may reflect stimulus responses [17]. The occurrence of an SCR is expected after a stimulus is applied [18].

SCR amplitude, rise time, decay time, and peak are the major parameters describing the shape of an SCR. These measures are also widely adopted as indicators of attention and information processing [19]. The presence of an SCR indicates that a novel or significant stimulus has been detected for further control processing. SCR amplitudes are usually interpreted as measures of attention [19]. Both the area under the curve (AUC) and mean amplitude are effective quantifications of the underlying autonomic arousal [20]. The first derivative, which passes the zero line if the curve shows maxima or minima, was recommended to obtain the time point of an SCR peak [21]. The SCR peaks duration accounts for the SCR peaks' mean rise and recovery time [22]. The SCR rise time is relatively seldom studied, and its relationship to psychophysiological processes is poorly understood [23,24]. Luong and Holz [22] found that the rise time of the skin conductance response (SCR) exhibited a comparable increasing trend to that of the SCR peak interval, while Langner et al. [25] noted a comparable increase in SCR rise time in association with amplitude.

Non-hairy (glabrous) skin covers heat-exchange organs; its temperature is not an ideal measure of ambient temperature but is more important for driving most thermoregulatory behaviors [26]. Local sweating rate is directly related to the instantaneous value of the skin

temperature during skin heating [27]. An association exists between the nature of the incentive and the temperature of the hands [28]. Compared with EDA, the skin temperature is more sensitive to emotional valence [28].

Movement is commonly recognized as a source of artifacts in the EDA signal. A conflict between expected and actual interactions with the surroundings (i.e., a mismatch between vestibular and visual signals) may cause motion sickness [29]. The symptoms include arousal change (drowsiness, yawning, lethargy, weariness), nausea and/or vomiting, stomach awareness and/or pain, increased autonomic activity (cold sweating, pallor, flushing, fluctuation in body temperatures), dizziness, and/or headache [30]. Additionally, there is no coherent or reliable relationship between these physiological measures (i.e., EDA and skin temperature) and subjective motion sickness due to external factors such as emotional states and environmental conditions [31].

2.2. Relationship between EDA and user experiences in VR games

Most prior studies focused on using physiological data to assess players' VR in-game experiences. It seems a high degree of emotion can increase the average number of SCRs. Bender and Sung [32] found that players' enjoyment was correlated with the average number of SCR in a VR killing zombies study. Luong and Holz [22] reported that higher frustration levels significantly increased the number of SCR, decreased the SCR peak amplitudes, shortened the time it took for peaks to rise, and shortened the time it took for SCR peaks to recover in VR. However, Navarro et al. [33] found no strong correlation between the number of SCR per minute and the evaluated experience variables in the Game Experience Questionnaire (i.e., GEQ [7]) in VR games. They explained that the variations in the EDA might display a stronger relationship with the differences in the player experience than the variations in SCR per minute.

Additionally, it seems that the level of EDA is determined by the mental effort needed in VR games. Egan et al. [34] found that the mean EDA value was positively correlated with participants' score for the item "needed to learn a lot of things before I could get going with this system" in a VR forklift driving task. Furthermore, an optimized game experience has the potential to reduce players' EDA levels during gameplay. Nacke et al. [35] found a negative relationship between flow and EDA when sound was on and between immersion and EDA when music was on in a desktop first-person shooter (FPS) game study. They inferred that flow and immersion could be more likely related to mental concentration effort in games (naturally leading to lower EDA). However, it is important to note that a lower EDA value can also be associated with the intensity of negative emotions in certain contexts. Drachen et al. [36] reported that the mean EDA value for each 5-minute segment was correlated with a negative emotion (i.e., possibly frustration) in desktop FPS video games.

2.3. Using ML models for assessing/predicting emotions

In addition to using traditional quantitative analysis to provide insights into the autonomic nervous system and affective states, researchers started to use ML algorithms to recognize players' emotions in various scenarios [37]. Their research shows the high potential of using ML to recognize players' stress [38,39], cognitive load [40], arousal [41], anxiety [42], and fear and frustration [22].

Specific to EDA in VR environments, it seems EDA features are among the most important physiological features in modeling players' emotions in VR. Granato et al. [41] reported that the level of EDA is one of the most informative signals for assessing a player's emotional intensity (from "very calm" to "very excited") in a VR race game among EDA, heart rate, electromyography, and respiration rate within various ML models. They explained that this might be due to the demand for maintaining attention and increasing stress levels in the game. However, there are exceptions. Orozco-Mora et al. [39] provided an assessment of

stress level estimation from ECG, EDA, and EMG signals while users played a VR video game with different difficulty levels. An 83.1 % accuracy was obtained for classifying between the three levels with a k-nearest neighbors (KNN) model. No significant statistical differences between the three difficulty levels were found by analyzing the EDA features. Collins et al. [40] attempted to use EDA features to recognize three levels of cognitive load in a VR learning context. They found that the most successful classifier was the Random Forest (RF) approach, which achieved about 51 % accuracy. They explain the low accuracy because the EDA takes longer to stabilize in a complex, changing environment.

Furthermore, skin temperature is known to be regulated by the autonomic nervous system. Thus, though its importance is less than EDA, skin temperature can contribute to the prediction modeling. Cho et al. [38] reported that a high accuracy of five stress levels (>95 %) could be estimated with the integrated features from EDA, skin temperature, and blood volume pulse by a novel ML algorithm. EDA provided more information about the stress level than the skin temperature. The mean of SCL, maximum of SCR, and mean of skin temperature are among the features with significantly different values between stress levels. Šalkevičius et al. [42] used patients' EDA, skin temperature, and blood volume pulse signals in VR exposure therapy to predict the anxiety level for assessing various anxiety disorders. Their four-level anxiety model can reach 86 % accuracy with a support vector machine (SVM) classifier. When these signals were used to predict anxiety, EDA had higher accuracy than the remaining two.

2.4. DPDR after VR gaming

The sudden removal of a VR headset may induce mental disorders in some VR players. Specifically, certain players might feel a sense of estrangement and unreality concerning their own self, a condition known as depersonalization (DP), whereas others may perceive a detachment or un reality from their environment, a state referred to as derealization (DR) [43–45]. Currently, research on DPDR experiences following VR gaming is rare, with only two studies identified. Barreda-Ángeles and Hartmann [43] reported that DPDR are more prevalent among novice players and among those who engage VR in longer sessions. Peckmann et al. [44] found that VR-action gaming can induce higher levels of DPDR experiences compared to classical personal computer (PC) gaming, though these effects are transient. They inferred that this phenomenon is linked to players' motion sickness experienced in VR.

In contrast to the limited research conducted on DPDR after VR gaming, prior studies have extensively explored motion sickness in VR. Players' motion sickness in VR is closely associated with the VR content and players' locomotion type [46,47]. Notably, content involving active interaction (e.g., shooting) and visual movement are reasons leading to a higher possibility of motion sickness in VR. In addition, prior research has underscored the potential of employing ML models that integrate action/movement data with other physiological metrics to forecast players' inclination to motion sickness in VR. For example, these action/movement data can be the length of the virtual path that an avatar traveled in a virtual-navigation task [48] and the head trajectory length in a VR driving game [49]. Additionally, EDA and skin temperature data are also demonstrated to be valuable in classification models for detecting motion sickness in VR [31,50–54].

2.5. Summary and research questions

Researchers have extensively investigated EDA and the relationship between EDA measures and central nervous system processes, as well as insights into skin temperature and movement. They have also identified the possible link between these EDA measures and players' experiences in a game. Notably, players' mental exertion positively affects their EDA values during gaming, while both an optimized gaming experience and

negative emotions can lower players EDA levels in the game.

Furthermore, ML algorithms are increasingly used to model players' emotions in VR, with EDA as a crucial feature. Skin temperature, though less significant, also aids prediction. Classification models, using physiological, skin temperature, and action data, can detect VR motion sickness with high accuracy. However, few studies have explored the potential association between in-game physiological data and players' post-game experiences. Therefore, we propose the following research questions:

Research Question (RQ) 1: What is the relationship between the various in-game EDA measures and players' post-game experiences?

RQ2: Can we use the in-game physiological data to predict players' post-game experiences through ML modeling?

Furthermore, Peckmann et al. [44] proposed that the DPDR experience after VR was related to players' motion sickness in VR. Given that action is a key factor contributing to motion sickness in VR, we formulated the third research question:

RQ3: Does players' in-game action cause their DPDR experience after taking off their VR headsets?

3. Methods

3.1. Participants

The study was submitted to a regional ethics board in China for its evaluation and was granted ethical approval. We recruited 64 participants from a large public university in Beijing, China. Table 1 summarizes the participants' demographic information collected through single-choice questions in the online registration survey. Notably, men had richer PC game experience than women in general and the difference is statistically significant: $\chi^2(4) = 15.92, p = 0.03$.

3.2. Materials

3.2.1. VR game condition

The experiment was conducted within a designated 3 m x 4 m entertainment area, utilizing a Vive Pro system loaded with the critically acclaimed single-player VR First-Person Shooter (FPS) game, Superhot (See Fig. 1). We selected this game in our experiment because the game plot is thrilling, exciting, and involves actions. It is conducive to stimulating players' diverse gaming experiences and facilitates us in detecting the potential influence of actions on post-VR game DPDR experience. This immersive game, developed by the eponymous team, is renowned for its innovative gameplay mechanic, where time progresses solely with the player's movements. This unique feature creates a

Table 1
Demographic information for study participants.

	<i>n</i>	%
Gender		
Men	44	69
Women	20	31
Age		
Between 18 and 20 years old	3	5
Between 20 and 25 years old	46	72
Between 25 and 30 years old	10	16
Between 30 and 40 years old	5	7
PC game experience		
Less than 1 year	10	16
Between 1 and 2 years	4	6
Between 2 and 5 years	22	34
Between 5 and 10 years	22	34
More than 10 years	6	10
VR game experience		
Less than 1 year	43	67
Between 1 and 2 years	13	20
Between 2 and 5 years	6	10
Between 5 and 10 years	2	3



(a)



(b)

Fig. 1. A fighting scenario in the VR game. (a) The VR fighting scenario; (b) A player of this VR game.

strategic and intense combat environment, requiring players to carefully plan their every action and movement to effectively navigate through levels teeming with enemy assaults. Players assume the role of a time-bending protagonist, using this extraordinary ability to evade bullets, outsmart foes, and neutralize threats. As they progress through the game's narrative, they encounter a diverse array of enemy gunmen and are introduced to new weaponry and adversaries with unique abilities and tactics, challenging their reflexes and strategic thinking. The entire experience was powered by a high-performance personal computer, ensuring smooth and seamless gameplay.

3.2.2. Self-reported measures

The post-game module of GEQ (i.e., pGEQ) [7] was used to gather information from players' self-reported post-game affective experiences. The GEQ is one of the most commonly used questionnaires for assessing players' video game experiences across countries [55]. As best as we know, the pGEQ stands out as the only popular questionnaire specifically tailored to assess post-game experiences. Other renowned game experience scales, such as the Player Experience of Need Satisfaction questionnaire [56], the Game Engagement Questionnaire [57], and the game user experience satisfaction scale [58], primarily focus on in-game emotions and measure aspects such as flow, presence, and immersion. The pGEQ comprises 17 items, covering a variety of distinct emotions that players might experience upon completing a gaming session. These items are categorized into four dimensions: positive experience, negative experience, tiredness, and returning to reality. In line with previous findings that highlighted the instability of the in-game GEQ (i.e., iGEQ) validity [55], the data gathered in our study did not align cleanly with the four predetermined categories of the pGEQ or any of the 2–6 factor solutions derived from exploratory factor analysis conducted using SPSS 27 (see Supplementary Material, Exploratory Factor Analysis of the pGEQ). However, this limitation did not hinder our item-by-item analysis of the questionnaire. The pGEQ employs a 5-point Likert scale, ranging from “strongly disagree” to “strongly agree.” Thanks to the nuanced nature of the pGEQ items, we were able to precisely differentiate between the various emotions experienced by players after gaming. Particularly, we utilized items in the returning to reality dimension to measure participants' DPDR experience after wearing off their VR headsets in this study.

3.2.3. Physiological measures

An E4 wristband [15] was used to collect players' physiological data in the game. This wristband can collect players' heart rate (including blood volume pulse and heart rate variation), EDA, skin temperature, and movement acceleration data simultaneously. The sampling frequency is 4 Hz in this study. However, due to the significant arm movements involved in the FPS game and the resulting decrease in

accuracy of heart rate measurements obtained from wearables during physical activity as compared to rest [59], the present study only targeted EDA, skin temperature, and movement acceleration (representing player actions) for analysis.

3.3. Procedure

To ensure a seamless transition, participants entered the laboratory in sequential order, with each individual following the departure of the preceding one. Upon arrival in the laboratory, an experimenter introduced the experiment and briefed them on their right to terminate the experiment at any point if they felt discomfort. Additionally, they were required to complete and sign an informed consent form. As part of their preparation, the experimenter led these participants through the game's desktop version. Afterwards, these participants put on the E4 wristband on their dominant hand and began engaging in the VR game. Participants were instructed to engage in the game across five separate 5-minute sessions, each representing a distinct VR fighting scenario. This structured approach ensured researchers to obtain 5 paired self-reported and physiological data per participant under the circumstances of limited human resources and funds. Specifically, once a game session was completed, the participant removed the VR headset and completed the pGEQ without distraction. Following a 5-minute break in the same location, they resumed the VR experience for the next session. Successful completion of all five sessions, which entailed engaging gameplay and responses to the questionnaires, was rewarded with a monetary incentive. Upon conclusion, participants exited the laboratory. Fig. 2 shows the experimental flow.

3.4. Data analytical approach

We used the Python programming system (version 3.10.6) for data analysis. The core packages used were Numpy (version 1.23.2) and Pandas (version 1.4.3) for data processing, Imblearn (version 0.11) for the oversampling of imbalanced data, Scikit-learn (version 1.1.2) for ML analyses, and Mlxtend (version 0.21.0) for feature selection analyses. Additionally, the (algorithmic) codes from an open-source program named EDA-Explorer from the MIT media lab [60] were partially used to process physiological data.

3.4.1. Physiological data preprocessing

Each participant's physiological data record was divided into 10 segments, which corresponded to their five game sessions in the VR environment and five questionnaire-filling timeslots in the real world. Only the physiological data for game sessions were processed in the following analyses. First, a first-order Butterworth low-pass filter with a cut-off frequency of 5 Hz was employed to remove noise from the

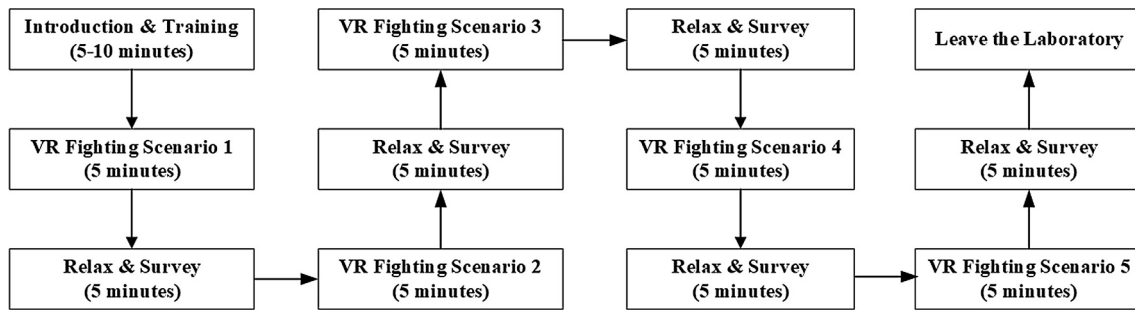


Fig. 2. Experimental flow.

physiological data. Then, Taylor et al.'s [60] algorithms were used to detect the EDA signal peaks and players' step computations. The parameters for the detection of EDA phase signals were as follows: (1) minimum amplitude = 0.02 μ S; (2) maximum rise time = 4 s; (3) offset = 1 s; and (4) maximum decay time = 4 s. Using the preprocessed physiological data, we developed 94 features, which were categorized into the following six groups:

- 1) Gender, age, PC game experience, VR game experience, session number.
- 2) The mean, maximum, minimum, median, range, SD, skewness, and kurtosis of the EDA signal.
- 3) The maximum, minimum, maximum of absolute value, and mean absolute value of the first and second differences of the EDA signal.
- 4) The mean, maximum, minimum, median, range, and SD of the height of EDA peaks; the interval of EDA peaks; the rise time of EDA peaks; the maximum derivation of EDA peaks; the amplitude of EDA peaks; the decay time of EDA peaks; the SCR width of EDA peaks; and the AUC of EDA peaks.
- 5) The mean, maximum, minimum, median, range, SD, skewness, and kurtosis of skin temperature.
- 6) The mean, maximum, minimum, median, range, SD, skewness, and kurtosis of step time during movement; stillness (%); and step count.

3.4.2. Correlation and ML analyses

After preprocessing the collected physiological data, we answered the proposed three research questions by steps. To address RQ1, which focuses on the relationship between various in-game EDA measures and players' post-game experiences, we performed Pearson correlation analyses. These analyses were conducted to assess the strength of the association between players' self-reported scores and mean EDA measures (e.g., the mean EDA, mean interval of the EDA peak, mean EDA peak, and mean rise time of SCR). We then identified and interpret patterns, such as the direction of the correlation and whether it was statistically significant, between these measures and the items in the pGEQ.

To address RQ2, which explores the possibility of predicting players' post-game experiences through ML modeling of in-game physiological data, we attempted to develop ML models for each item utilizing the RF, KNN, multilayer perceptron (MLP), and SVM algorithms. These classification algorithms were chosen due to their widespread adoption in state-of-the-art emotion classification research [39]. The amount of data at the five levels was imbalanced for some items. For example, the first level was only selected six times in Item 8 ("I felt satisfied"). Thus, we employed the SMOTE algorithm [61] to balance the data (i.e., added samples of the minority class) to equality; thus, classifiers were prevented from being biased toward the majority class for each item prior to modeling. Models were built with the algorithms' default values, and the results were validated with 5-fold cross-validation. These models' prediction accuracy and f1 score were reported. Notably, cross-validation doesn't explicitly split data into training and testing sets; instead, it automatically partitions the dataset into 5 subsets (or "folds"). In each iteration, 4 subsets are used for training and 1 for validation, with this

process repeating 5 times so that each subset serves as the validation set once. This method is more robust than percentage split for evaluating model performance as it mitigates bias from improper data division [62].

Finally, to investigate whether players' in-game actions contribute to their experience of DPDR after removing their VR headsets, as posed in RQ3, we employed RF to identify the 10 most important features for items in the returning to reality dimension of the pGEQ. The reason for choosing items in the returning to reality dimension is because these items (items 3: "I found it hard to get back to reality"; 9: "I felt disoriented"; and 17: "I had a sense that I had returned from a journey") show representative phenomena of the symptoms defined in DPDR (experiencing sensations of unreality and feelings of detachment from the physical surroundings or of one's own body [45]). We used the RF feature importance method to remove redundant features that have high correlations with each other, and stronger relationships are indicated by higher index values [63]. Then, we calculated the distribution of the top 10 most important features for items in the returning to reality dimension across feature groups. This calculation helped to determine the general association between action features and players' ratings for items related to their experience of DPDR after taking off their VR headsets.

4. Results

In this section, we first provide a brief descriptive data report and then present the results of our data analysis following the steps outlined in Section 3.2 ("data analytical approach"). After presenting the results in Section 4, we compare our findings with prior research and explore new insights in Section 5 ("Discussion").

4.1. Descriptive statistics and repeated-measure tests

Table 2 provides an overview of the mean scores for items in the pGEQ across five sessions. Among the 17 items, Item 8 ("I felt satisfied") had the highest mean scores, while Item 14 ("I felt regret") and Item 15 ("I felt ashamed") had the lowest mean scores consistently across all sessions. Furthermore, we conducted a series of repeated-measures ANOVA tests to determine whether there were any significant differences in the ratings of these items across the five scenarios. The p-values for the comparisons of these 17 items ranged from 0.14 to 0.99, indicating that there were no significant differences in the scores for any item among the different scenarios.

4.2. Correlation between items and mean EDA measures

Table 3 presents the Pearson correlations between subjective ratings, PC game experience, VR game experience, and the mean EDA measures by categories, following the removal of noise data. Overall, the number of significant correlations between a mean EDA measure and items in the pGEQ ranges from four to nine. Additionally, we have discerned five notable patterns:

Table 2

Mean scores for items in the pGEQ.

No.	Item	Session 1		Session 2		Session 3		Session 4		Session 5	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1	I felt revived	3.97	1.04	4.06	0.97	3.92	0.93	4.05	0.95	3.98	0.95
2	I felt bad	1.77	0.79	1.89	0.93	1.77	0.73	1.83	0.90	1.77	0.71
3	I found it hard to get back to reality	1.95	0.95	2.05	1.05	1.95	1.00	2.02	0.92	2.02	1.02
4	I felt guilty	1.77	0.83	1.77	0.96	1.70	0.79	1.70	0.85	1.67	0.78
5	It felt like a victory	3.55	1.01	3.61	0.95	3.67	1.04	3.45	0.96	3.48	1.05
6	I found it a waste of time	1.78	0.84	1.70	0.77	1.75	0.80	1.72	0.86	1.73	0.74
7	I felt energized	3.77	1.02	3.75	0.98	3.69	0.99	3.77	0.99	3.73	0.98
8	I felt satisfied	4.13	0.75	4.22	0.81	4.08	0.84	4.13	0.81	4.06	0.89
9	I felt disoriented	1.72	0.77	1.66	0.70	1.81	0.85	1.69	0.69	1.81	0.81
10	I felt exhausted	2.27	1.09	2.20	1.13	2.28	1.21	2.33	1.11	2.20	1.07
11	I felt that I could have done more useful things	2.03	0.89	1.94	0.69	2.02	0.83	1.92	0.84	1.97	0.84
12	I felt powerful	3.11	1.07	3.11	1.16	3.11	1.16	3.14	1.13	3.22	1.27
13	I felt weary	2.17	0.98	2.19	0.96	2.02	0.92	2.16	0.93	2.06	0.96
14	I felt regret	1.59	0.61	1.63	0.58	1.66	0.57	1.56	0.59	1.53	0.53
15	I felt ashamed	1.61	0.58	1.61	0.55	1.56	0.56	1.67	0.59	1.56	0.56
16	I felt proud	3.39	1.23	3.36	1.16	3.31	1.19	3.31	1.10	3.23	1.11
17	I had a sense that I had returned from a journey	3.70	1.05	3.75	1.04	3.72	0.98	3.72	0.98	3.72	1.00

Table 3

Pearson correlation between the subjective ratings, prior game experience, and the mean EDA measures.

No.	Item ^Δ	PC game experience	VR game experience	Mean	Mean peak	Mean peak interval	Mean rise time	Mean delay time	Mean amplitudes	Mean decay time	Mean SCR duration	Mean AUC
Positive experience												
1	I felt revived	0.13*	0.16**	-0.01	0.01	0.15**	0.12*	-0.08	-0.07	-0.03	-0.05	-0.06
5	It felt like a victory	0.17**	0.10	0.11	0.10	0.12*	-0.01	0.07	0.04	-0.12*	-0.12*	0.01
7	I felt energized	0.39**	< 0.01	0.03	0.01	0.12*	-0.02	0.03	0.01	0.02	0.02	0.02
8	I felt satisfied	0.11	-0.07	0.16**	0.12*	0.08	-0.01	0.17**	0.15**	-0.07	-0.03	0.15**
12	I felt powerful	0.09	0.12*	0.22**	0.21**	0.23**	0.06	0.02	0.04	-0.08	-0.07	0.05
16	I felt proud	0.17**	0.01	0.24**	0.25**	-0.06	-0.04	0.27**	0.30**	-0.19**	-0.20**	0.27**
Negative experience												
2	I felt bad	-0.10	-0.04	-0.12*	-0.12*	0.02	0.02	0.02	< 0.01	0.21**	0.23**	-0.02
4	I felt guilty	-0.04	0.03	-0.11*	-0.08	0.09	-0.08	0.08	0.11	-0.07	-0.07	0.10
6	I found it a waste of time	-0.10	-0.12*	-0.23**	-0.20**	0.08	0.16**	-0.17**	-0.13*	0.12*	0.07	-0.15**
11	I felt that I could have done more useful things	-0.10	-0.22**	-0.16**	-0.16**	0.13*	0.14*	-0.12*	-0.04	0.10	0.12*	-0.05
14	I felt regret	-0.28**	-0.08	-0.08	-0.09	-0.07	0.09	-0.07	-0.05	0.02	0.03	-0.06
15	I felt ashamed	-0.29**	-0.03	-0.15**	-0.14*	0.05	0.10	-0.15**	-0.13*	0.05	0.03	-0.14*
Tiredness												
10	I felt exhausted	-0.17**	-0.19**	0.09	0.08	-0.25**	-0.10	0.18**	0.06	-0.09	-0.06	0.01
13	I felt weary	-0.19**	-0.24**	-0.01	0.01	-0.01	0.17**	-0.07	-0.03	0.08	0.08	-0.04
Returning to reality												
3	I found it hard to get back to reality	-0.37**	0.04	-0.11*	-0.12*	0.04	0.03	0.02	0.03	0.05	0.11	0.04
9	I felt disoriented	-0.09	-0.06	-0.05	-0.04	0.08	0.01	0.15**	0.14*	-0.05	-0.01	0.12*
17	I had a sense that I had returned from a journey	-0.01	-0.33**	0.11	0.12*	-0.06	0.04	0.10	0.10	-0.03	-0.07	0.12*
Number of significant correlations between a mean EDA measure and items in the pGEQ												
				9	9	6	4	7	5	4	4	6

Note. * $p < 0.05$; ** $p < 0.01$; ^Δ 5-point Likert scale.

First, five in six items belonging to positive experiences had a significant positive correlation with either prior PC or VR game experience, while four in six items belonging to negative experiences had a significant negative correlation with either prior PC or VR game experience. Second, among the EDA measures, the mean, mean peak, and mean interval of EDA exhibited the highest number of significant positive correlations with items in the positive experience category, while the mean and mean peak of EDA showed the highest number of significant negative correlations with items in the negative experience category.

Third, when considering only significant correlations, we found that most EDA measures demonstrated a different direction of correlation with items in the positive experience category versus those in the negative experience category. For example, the mean EDA delay time was positive correlation with Items 8 and 16 in the positive experience category while negative correlation with Items 6, 11, and 15. The only exceptions were the mean peak interval and the mean peak rise time, both of which showed a positive correlation with items in the positive experience category and items in the negative experience category.

Fourth, the items related to tiredness demonstrated a significant negative correlation with prior PC and VR game experience. This pattern closely resembles the pattern seen in items belonging to the negative experience dimension. Consequently, we conducted a correlation analysis between the items in the tiredness dimension and those in the negative experience dimension (See Table 4). The results revealed a positive correlation between the items in the tiredness dimension and those in the negative experience dimension.

Finally, the relationship between the mean EDA measures and the three items belonging to the returning to reality dimension does not exhibit any clear patterns. To gain deeper insights, we further explored the players' explanations for their selections of these items and found that they perceived them from diverse perspectives. Most players viewed Item 3 ("I found it hard to get back to reality") as a negative experience, a sentiment they did not share. For example, one player said, "The short-term experience didn't make me feel lost." Another one said, "It's a great way to relax and also exercise the body." In contrast, players regarded Item 17 ("I had a sense that I had returned from a journey") as a positive experience. One player explained their choice by saying, "There's a feeling of traveling through time and space, which is truly wonderful." Differing from their perspectives on Items 3 and 17, players' responses to Item 9 ("I felt disoriented") were closely tied to their feelings when transitioning from the virtual world to the physical world. One player shared, "I can return to reality and distinguish directions, among other things," while another noted, "I can return to normal immediately after leaving the game."

4.3. Models predicting post-VR game experiences

Following the steps outlined in Section 3.4. ("Data analytical approach"), we extracted and then classified 94 features into six distinct groups:

- Group 1: Features related to the demographic information.
- Group 2: Features related to the mean EDA signal.
- Group 3: Features related to the changing rate of EDA signal.
- Group 4: Features related to other EDA signals.
- Group 5: Features related to skin temperature signals.
- Group 6: Features related to action signals.

We subsequently employed five-fold cross-validation to structure the modeling and evaluation processes. The MLP, RF, KNN, and SVM algorithms, with their default parameters, were utilized to independently construct prediction models for each item.

The study's results revealed that the data gathered from the wrist-band possessed the potential to predict users' post-VR game experiences. Table 5 presents the prediction accuracy of the established model for each item in the pGEQ with the selected four algorithms. In general, the models built with the MLP algorithm had the highest average accuracy

Table 4

Pearson correlation between items in the tiredness dimension and items in the negative experience dimension.

		Tiredness Item10: I felt exhausted	Item13: I felt weary
Negative experience	Item2: I felt bad	0.25**	0.29**
	Item4: I felt guilty	0.17**	0.34**
	Item6: I found it a waste of time	0.26**	0.33**
	Item11: I felt that I could have done more useful things.	-0.06	0.59**
	Item14: I felt regret.	0.29**	0.38**
	Item15: I felt ashamed.	0.27**	0.34**

Note. * $p < 0.05$; ** $p < 0.01$.

and f1 score. However, no items in the positive experience category surpassed an accuracy of 70.00 %. Considering this limitation, we recategorized players' subjective ratings into three levels: "disagree" (scores 1 and 2), "neither disagree/agree" (score 3), and "agree" (scores 4 and 5). We then developed prediction models using the same physiological data for these three rating levels.

Table 6 displays the prediction outcomes for players' post-VR gaming experiences utilizing the refined 3-point scale. The models based on the RF and the MLP algorithms achieved the highest average accuracies, at 79.98 % and 79.03 % respectively, demonstrating average improvements of 16.77 % and 12.50 % compared to the results obtained using the 5-point scale. Notably, with the MLP model, half of the items in the positive experience category achieved an accuracy exceeding 70.00 %.

4.4. Actions in VR and post-game DPDR experience

Aligned with our analyses in establishing ML models for predicting post-VR game experiences, we classified all 94 features into six distinct groups. Then we employed RF to identify the 10 most important features for items in the returning to reality dimension of the pGEQ. Table 7 displays the count of most important features within each feature group for the items related to returning to reality. In general, items in the returning to reality dimension shows a close relationship with features related to actions (i.e., feature in Group 6) in their prediction models, respectively. The existence of this close relationship does not depend on whether or not Group 1 features (features related to players' demographic information) are involved.

Furthermore, PC game experience (in Group 1) is in the top 10 important features for Items 3 ("I found it hard to get back to reality"). VR game experience (in Group 1) is in the top 5 for Item 17 ("I had a sense that I had returned from a journey"). However, no features in Group 1 are ranked in top 20 for Item 9 ("I felt disoriented").

5. Discussion

This study explored the potential of utilizing data from wearable physiological sensors to predict diverse post-VR gaming experiences. The key areas of focus included: identifying the relationships between EDA measures and subjective evaluations, developing predictive models, and investigating the potential for post-VR game DPDR linked to players' in-game actions. To the best of our knowledge, this is the first to examine the connection between in-game psychological data and players' post-game experiences.

This research makes four significant contributions to the field of wearable sensors and VR gaming experiences. Firstly, it establishes that EDA measurements, particularly the mean EDA and mean EDA peak, has the potential to serve as valuable indicators of players' post-VR gaming experiences, after noise data has been filtered out. Secondly, the study indicated that 3-level post-VR gaming experiences can be predicted with much higher accuracy than 5-level ones by utilizing data from wearable physiological sensors combined with a ML method. Third, it reveals that players' DPDR after the VR game is influenced by the intensity of their actions in VR. Prior game experience impacts DPDR, but it cannot account for all observed phenomena. Lastly, we verified the instability of the factor structure in the pGEQ and emphasized that this issue stems from the overlap among the four pre-defined categories. These four contributions support future VR game centers to remotely analyze player loyalty at a low cost. This accomplishment establishes the necessary foundation for these centers to generate customized new 3D game scenarios, thereby enhancing players' post-game experiences with the assistance of future advanced generative artificial intelligence technologies.

Table 5Predicting players' post-VR game experiences using a 5-point Likert scale ^Δ.

No.	Item	MLP (max iteration = 1000)		RF		SVC		KNN	
		accuracy (%)	f1_score (%)	accuracy (%)	f1_score (%)	accuracy (%)	f1_score (%)	accuracy (%)	f1_score (%)
Positive experience									
1	I felt revived	65.50	62.96	63.50	60.38	61.50	59.08	59.17	53.39
5	It felt like a victory	65.54	62.03	62.14	58.46	57.50	54.73	56.61	50.95
7	I felt energized	64.07	61.30	61.11	58.32	55.00	52.27	51.48	47.71
8	I felt satisfied	54.88	50.97	53.86	49.81	48.97	46.88	46.35	40.15
12	I felt powerful	53.64	52.27	50.68	45.67	50.23	46.07	47.50	40.62
16	I felt proud	59.14	58.39	57.63	54.92	52.26	48.98	47.31	43.04
Negative experience									
2	I felt bad	72.03	66.65	70.62	67.92	60.65	57.27	65.55	61.87
4	I felt guilty	77.62	76.46	78.46	77.41	77.90	77.01	70.07	66.99
6	I found it a waste of time	75.17	73.41	70.86	69.51	68.10	65.48	66.03	63.65
11	I felt that I could have done more useful things	81.68	76.50	81.83	79.55	81.99	80.84	78.78	75.77
14	I felt regret	68.54	64.92	73.33	70.49	69.79	68.41	57.92	54.48
15	I felt ashamed	75.50	72.26	72.49	69.55	73.30	71.98	66.47	63.36
Tiredness									
10	I felt exhausted	82.26	80.15	81.76	79.63	76.35	74.13	74.84	70.19
13	I felt weary	84.71	80.90	80.51	77.42	82.80	81.86	80.89	78.10
Returning to reality									
3	I found it hard to get back to reality	85.59	83.06	81.49	78.95	84.47	82.82	79.63	76.44
9	I felt disoriented	66.62	61.40	63.48	59.50	63.66	61.62	57.00	51.57
17	I had a sense that I had returned from a journey	61.73	60.44	60.70	58.24	53.11	50.77	53.31	49.99
Average		70.25	67.30	68.50	65.63	65.74	63.54	62.29	58.13

Note. ^Δ: Models were built with the algorithms' default parameter values, and the results were validated with five-fold cross-validation.**Table 6**Predicting players' post-VR game experiences using the revised 3-point scale ^Δ.

Item No.	MLP (max iteration = 1000)				RF				SVC				KNN			
	Accuracy (%)		f1_score (%)		Accuracy (%)		f1_score (%)		Accuracy (%)		f1_score (%)		Accuracy (%)		f1_score (%)	
	value	boost °	value	boost °	value	boost °	value	boost °	value	boost °	value	boost °	value	boost °	value	boost °
Positive experience																
1	77.23	17.90	74.84	18.86	76.64	20.70	74.51	23.42	79.02	28.49	77.59	31.33	69.21	16.97	62.50	17.07
5	65.10	-0.67	57.96	-6.56	60.59	-2.50	57.19	-2.17	56.86	-1.11	54.11	-1.13	55.49	-1.97	49.92	-2.02
7	72.79	13.60	70.92	15.69	68.25	11.67	64.17	10.03	64.41	17.11	60.39	15.54	64.58	25.45	58.10	21.77
8	74.57	35.86	71.73	40.73	76.63	42.27	74.44	49.46	71.79	46.60	69.53	48.32	67.50	45.64	59.88	49.14
12	56.69	5.70	55.25	5.70	52.49	3.57	50.76	11.14	48.04	-4.36	47.51	3.13	48.81	2.75	46.44	14.33
16	60.78	2.77	53.37	-8.60	57.22	-0.72	52.30	-4.78	48.25	-7.68	44.46	-9.22	52.97	11.97	46.59	8.24
Negative experience																
2	86.30	19.81	84.11	26.19	91.11	29.02	90.92	33.85	85.80	41.46	85.10	48.60	75.06	14.50	70.40	13.77
4	88.93	14.57	88.82	16.17	94.18	20.03	94.13	21.60	92.78	19.10	92.59	20.22	75.06	7.12	70.34	4.99
6	84.45	12.34	83.06	13.14	85.20	20.24	84.60	21.71	86.43	26.91	85.85	31.12	72.26	9.43	66.82	4.98
11	91.12	11.56	90.28	18.01	91.38	11.67	91.18	14.63	90.87	10.83	90.63	12.12	79.15	0.47	76.76	1.31
14	86.78	26.61	88.72	36.67	97.69	33.21	97.68	38.59	93.55	34.05	93.50	36.68	64.19	10.84	58.79	7.93
15	87.87	16.38	85.87	18.83	94.26	30.03	94.19	35.43	91.80	25.24	91.70	27.40	68.36	2.85	64.18	1.30
Tiredness																
10	82.19	-0.09	80.00	-0.19	88.61	8.38	88.38	10.99	77.92	2.06	76.87	3.70	73.22	-2.17	68.60	-2.27
13	82.55	-2.56	79.81	-1.36	80.74	0.28	79.29	2.42	81.29	-1.83	79.92	-2.37	73.70	-8.89	69.14	-11.47
Return to reality																
3	90.42	5.64	89.36	7.58	89.52	9.85	88.99	12.72	90.92	7.64	90.69	9.49	73.56	-7.62	68.45	-10.45
9	84.29	26.53	83.08	35.33	86.20	35.80	85.19	43.19	85.48	34.29	84.64	37.37	71.10	24.74	63.91	23.92
17	71.48	15.80	68.76	13.76	69.00	13.68	64.84	11.34	66.34	24.91	64.27	26.60	71.10	33.37	63.91	27.85
Average																
	79.03	12.50	76.82	14.15	79.98	16.77	78.40	19.45	77.15	17.36	75.85	19.37	67.96	9.10	62.63	7.74

Note. ^Δ: Models were built with the algorithms' default parameter values, and the results were validated with five-fold cross-validation. °: boost = (3-point value - 5-point value)*100/(5-point value).

5.1. Correlation between items and mean EDA measures

The mean and mean peak of EDA exhibit a significant positive correlation with most items related to positive experiences, and a negative correlation with most items related to negative experiences. This finding is consistent with the research conducted by Drachen et al. and the study by Mandryk and Atkins, both of which reported that a higher mean EDA is associated with positive emotions [36,64]. However, our results

diverge from those of Nacke et al., who found that immersion and flow can lead to lower EDA, without taking into account players' prior experience [35]. This discrepancy can be attributed to the fact that the level of EDA is influenced by the mental effort required in VR games [34]. More experienced players can more easily immerse themselves and maintain focus in the fighting scenario, whereas less experienced players may experience negative emotions due to their unfamiliarity with video games, even after a brief training session. This explanation also accounts

Table 7

The number of top important features in each feature group for the prediction model of items in “returning to reality”.

No.	Item [△]	Group 1		Group 2 [°]		Group 3 [°]		Group 4 [°]		Group 5 [°]		Group 6 [°]	
		top 5	top 10	top 5	top 10	top 5	top 10	top 5	top 10	top 5	top 10	top 5	top 10
Include Group 1 (Features related to demographic information)													
3	I found it hard to get back to reality	0	1	0	0	0	0	0	1	2	3	3	5
9	I felt disoriented	0	0	0	1	0	1	2	2	1	1	2	5
17	I had a sense that I had returned from a journey	1	1	2	2	0	1	1	1	1	2	0	3
Without Group 1 (Features related to demographic information)													
3	I found it hard to get back to reality			0	0	0	0	0	1	1	3	4	6
9	I felt disoriented			0	2	0	1	2	3	1	1	2	3
17	I had a sense that I had returned from a journey			1	1	0	1	1	3	1	2	2	3

Note. [△]: 5-point Likert scale.[°]: Group 2: Features related to the mean EDA signal. Group 3: Features related to the changing rate of EDA signal. Group 4: Features related to other EDA signals. Group 5: Features related to skin temperature signals. Group 6: Features related to actions.

for the significant negative correlation observed between tiredness-related items and the prior game experience. Specifically, while the game is thrilling and exciting for expert players, novices may have a less enjoyable experience and feel more fatigued.

Furthermore, consistent with previous literature, other key EDA metrics, including mean decay time, mean SCR duration, mean delay time, mean amplitudes, and mean AUC, are also valuable indicators of players' emotions [19,22,25]. When focusing solely on significant correlations, these measures demonstrated a different direction of correlation with items in the positive experience category versus those in the negative experience category. In contrast, we found that both the mean peak interval and the mean peak rise time exhibited a positive correlation with both items in the positive category and items in the negative experience category. This situation can be explained as that the direction and magnitude of the correlation between emotions and the mean peak interval are not stable, but rather are determined by the game genre [33], and that the mean peak interval accounts for the mean peak rise time [22].

5.2. ML models for post-VR game experiences

When comparing the prediction results in Tables 5 and 6, we observed that utilizing a 3-point scale significantly enhanced prediction accuracy and F1 scores, especially for items related to positive experiences. Therefore, the 3-point scale is more appropriate in scenarios where high prediction accuracy for positive experiences is crucial. The high accuracy of our established prediction models for post-VR game experiences aligns with findings from previous related studies, which have demonstrated that most 3-level models can achieve high accuracy in assessing emotions in VR games (i.e., [38,40,42]). However, the prediction in this study is distinctive from prior studies in its focus on players' post-VR game experiences, which more directly determine whether players accept the game than assessing players' in-game emotions and have more practical values in the VR game industry.

5.3. Actions features for predicting post-game DPDR

Peckmann et al. [44] found that VR-action gaming can induce higher levels of DPDR experiences than classical PC gaming. Our study confirmed that features related to players' in-game actions are among the most important predictors of their post-game DPDR. These results can be explained from the view of embodied cognition. That is, autonomous action allows players to have a higher level of agency (also called embodiment) [65], leading to a heightened immersion in the VR environment [66]. Upon returning to the physical world, players may experience a loss of agency and a sense of unreality due to prior immersion in the virtual context; however, they do not develop the false belief that they are no longer in control of their actions [67]. This perception conflict between the virtual and physical reality worlds leads to their DPDR after VR gaming. Our view point is aligned with

Limanowski and Friston's [68] view that DPDR arises from an updated high-level belief related to prior predictive coding.

Furthermore, our findings revealed that players' prior game experience can influence their susceptibility to DPDR after VR gaming. This result is aligned with the report by Barreda-Ángeles and Hartmann, which highlights a higher prevalence of DPDR among novice players [43]. However, our study also indicated that players' perception of disorientation is linked to their actions rather than their prior game experience. This result can be explained as that the disorientation after VR gaming may stem from changes in blood pressure, heart rate, inner ear, electrolytes, or neurological conditions that affect balance following intense actions [69]. The disorientation is not exclusive to post-VR scenarios but can occur in any condition involving physical activity. Therefore, the possibility and severity of disorientation after VR gaming is not influenced by players' prior game experience. Overall, our study results suggest that players' in-game action is a critical reason for their DPDR experience after removing VR headsets; while prior game experience impacts DPDR, it cannot account for all observed phenomena.

5.4. Structure validity of the GEQ

Law, Brühlmann, and Mekler [55] discovered that the factor structure of the in-game module of the GEQ (i.e., iGEQ) lacked stability following a confirmatory factor analysis of data collected through an online survey. Similarly, our study findings indicated that the items in the post-game module of the GEQ (i.e., pGEQ) failed to converge into the four pre-defined categories outlined in the questionnaire's introduction. The reason is that players' feelings of tiredness and returning to reality are linked to either positive or negative experiences. This is shown by the strong positive correlation between items in the tiredness and negative experience dimensions (see Table 4), and by how participants' responses to Items 3 and 17 can be categorized into negative and positive experience dimensions, respectively. In other words, the pre-defined four dimensions are not independent with each other. Consequently, traditional factor analysis methods are unsuitable for assessing the structure of the GEQ.

6. Limitations and future works

It is worth mentioning that there are potential limitations to this study. First, data used for the correlation analysis were collected from participants five times. A correlation likely exists between data points from the same participant in different sessions. Thus, we conducted correlation analysis for data in each session (see Supplementary Material, Pearson Correlation between the Subjective Ratings, Prior Game Experience, and the Mean EDA Measures across Sessions). Overall, although certain correlations presented in Table 2 are not significant in some sessions due to a smaller sample size, the direction of these correlations remained stable across each session in our analysis. In other words, the possible bias caused by the repeated measure is small.

Second, only one genre of VR games was assessed, and differences exist between different genres of VR games. Specifically, the VR FPS game used in this study emphasizes players' actions, while other VR games, such as meditation games, do not include intensive actions. The findings of this study should thus be validated in VR games that do not require intensive actions. Third, the only physiological signals explored in this study were EDA, skin temperature, and movement. Regarding EDA signals, we solely extracted features associated with SCR and excluded those pertaining to SCL. With the development of technology, researchers will likely soon be able to collect more types of physiological signals with more accurate values from a wristband. Thus, future predictions can be more accurate and demonstrate slight differences at the time.

Furthermore, other three areas remain to be explored in future studies. First, although some of the correlations presented in Table 2 achieved statistical significance ($p < 0.05$), the corresponding correlation coefficients suggest that these associations are relatively weak. To determine whether such weak relationships possess meaningful and practical values, further validation through more comprehensive or focused studies is required. Second, we have found that by categorizing players' agreement on their experiences into three levels, we can accurately predict their post-VR game experiences. Categorizing players' agreement into three levels is a common method for emotion prediction in VR studies [39,40,49]. Nevertheless, further investigation is required to ascertain whether this general categorization truly holds practical value for predicting and enhancing players' post-game experiences. Finally, there is a research gap regarding the duration of DPDR symptoms after VR gaming. While Peckmann et al. [44] found no significant DPDR differences between VR and PC groups from the second day after VR gaming, Barreda-Ángeles and Hartmann [43] reported more frequent DPDR symptoms among those with longer VR sessions. Our study found no significant differences in players' scores on returning-to-reality items across five 5-minute sessions with 5-minute intervals (See Section 4.1). However, future research should still investigate the duration of DPDR symptoms in post-VR gaming to determine whether DPDR is a mental issue needing high attention in VR studies.

7. Conclusions and implications

This study explored the viability of utilizing data from wearable physiological sensors to predict players' post-VR gaming experiences. Participants' EDA signals, skin temperature, and movement data were captured while they engaged in a VR FPS game. These datasets underwent processing for feature extraction. Through correlation analysis and the creation of classification models, we delved into the relationship between these features and the items of the pGEQ. The findings indicated that EDA features could be predictors of post-VR game experiences. Physiological metrics demonstrated high accuracy in forecasting players' 3-level post-VR gaming sentiments. Player actions influence their severity of DPDR after the VR session. While prior game experience impacts DPDR, it cannot account for all observed phenomena.

The study's insights lay the groundwork for a multimodal approach incorporating physiological data to predict and enhance user loyalty after playing VR games. For example, for a player experienced in a VR FPS game, if the ML models foresee a decline in the likelihood of positive post-game experiences based on real-time physiological data, introducing a new game scenario with more frequent actions can elevate the player's mean EDA and EDA peak values, thereby ensuring a consistently positive post-game experience. Conversely, a gentler and smoother game scenario would be more appropriate for novice players to engage in. Nevertheless, for all players, it is advisable to incorporate a calming scenario with minimal or no action towards the end of the game to prevent the onset of DPDR after VR gaming. These insights are important for the future VR gaming industry. They pave the way for creating real-time game scenarios that enhance post-game experiences, supported by advanced generative artificial intelligence technologies. It

is crucial to highlight, however, that the cornerstone of this accomplishment lies in obtaining the player's explicit consent to gather their physiological data through the wristband, thereby safeguarding against any potential encroachment on players' privacy.

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CRedit authorship contribution statement

Wen Huang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Jiayi Gao:** Writing – original draft. **Xinyuan Chen:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.entcom.2025.100977>.

Data availability

Data will be made available on request.

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